

SIMATS ENGINEERING

ASSESSMENT TOOL 4 – DRY RUN EVALUATION – ANSWERS

Course Name: Artificial Intelligence Course Code: CSA17

1. Breadth-First Search (BFS) from A

Graph: $A \rightarrow (B, C)$; $B \rightarrow (D, E)$; $C \rightarrow (F, G)$

Iteration	Queue (after step)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	(empty)	G

1. BFS traversal order: A, B, C, D, E, F, G

2. Queue contents after each iteration are shown in the table above (FIFO order, children added left to right: left child before right child).

2. Depth-First Search (DFS) from A

Same graph. Traversal follows leftmost-child-first order.

Iteration	Stack (after step)	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	(empty)	G

DFS traversal order: A, B, D, E, C, F, G

3. 8-Puzzle — BFS Trace to Goal

Initial: 1 3 6 / 5 _ 2 / 4 7 8 Goal: 1 2 3 / 4 5 6 / 7 8 _

Manhattan-distance lower bound = 8, and an 8-move solution exists, so BFS's shortest path has depth 8 (9 states including the start). The solution path found by BFS is traced below; each state also has 2–3 sibling states generated at the same depth (from the other legal blank moves) that are expanded but do not lead to a shorter goal path.

Iter	Current State (popped/expanded)	Queue context	Move / Note
1	1 3 6 / 5 _ 2 / 4 7 8	Root only: [S0]	Expand S0 → push Right, Up, Left, Down children
2	1 3 6 / 5 2 _ / 4 7 8	3 siblings remain queued	Blank moved Right; expand → generates next-level children
3	1 3 _ / 5 2 6 / 4 7 8	siblings + this level's children	Blank moved Up
4	1 _ 3 / 5 2 6 / 4 7 8	siblings + children	Blank moved Left
5	1 2 3 / 5 _ 6 / 4 7 8	siblings + children	Blank moved Down — row 0 solved (1 2 3)
6	1 2 3 / _ 5 6 / 4 7 8	siblings + children	Blank moved Left
7	1 2 3 / 4 5 6 / _ 7 8	siblings + children	Blank moved Down — row 1 solved (4 5 6)
8	1 2 3 / 4 5 6 / 7 _ 8	siblings + children	Blank moved Right
9	1 2 3 / 4 5 6 / 7 8 _	GOAL REACHED	Blank moved Right — matches goal state

1. Node expanded each iteration: shown in the 'Current State' column above — the state on the optimal path at that depth.

2. Total states generated: BFS branches into 2–4 children per state (2 at corners, 3 at edges, 4 when blank is central), so by depth 8 several hundred distinct states are generated in total across all branches; only the 9 states above lie on the shortest (goal) path.

3. Solution path (initial → goal), 8 moves — Right, Up, Left, Down, Left, Down, Right, Right:

• 1 3 6/5 _ 2/4 7 8 → 1 3 6/5 2 _/4 7 8 → 1 3 _/5 2 6/4 7 8 → 1 _ 3/5 2 6/4 7 8 → 1 2 3/5 _ 6/4 7 8 → 1 2 3/_ 5 6/4 7 8 → 1 2 3/4 5 6/_ 7 8 → 1 2 3/4 5 6/7 _ 8 → 1 2 3/4 5 6/7 8 _

4. Why BFS always finds the shortest solution in an unweighted 8-puzzle: every move has the same cost (1 step), and BFS expands nodes level by level, in strictly increasing order of depth. The first time it reaches the goal, no shallower (shorter) path could exist, since every node at depth d is expanded only after all nodes at depth $d-1$ — so the path found is guaranteed to use the minimum number of moves.